

Zone Submission – Media Forensics & Deepfake Lab

1. Zone Idea

Concept: A 3-hour hands-on forensic investigation lab where registered participants analyze manipulated media (images, video, audio) using real forensic tools. Participants solve **nine progressive challenges** (easy, medium, hard) across three sections:

- **ELA (Error Level Analysis)** – detect copy-move forgery in images.
- **De-blurring** – extract and sharpen frames to read a blurred license plate.
- **Spectrogram** – extract audio and find hidden messages (Morse code, DTMF, visual patterns).

Theme: *“You are the digital forensic analyst – uncover the truth hidden in pixels, frames, and sound.”*

Why it works:

- Structured 3-hour session ensures deep engagement.
- Real forensic techniques used by investigators.
- Immediate reward (flags, prizes) per solved challenge.
- Runs in registered batches to manage capacity.

2. Workflow

Before the event

- Prepare challenge files, tools, and HTTP server on zone laptop.
- Set up registration list (time slots: e.g., 9:00–12:00, 13:00–16:00, etc.).
- Print score sheets, challenge summaries, and prize tokens.

During a batch (3 hours)

Time	You Do	Participants Do
0:00–0:15	Intro, share IP, start HTTP server	Connect to Wi-Fi, download all files via browser
0:15–0:25	Explain Challenge 1 (ELA) + live demo	Listen, ask questions
0:25–1:15	Walk around, provide hints	Solve all 3 ELA challenges (easy → medium → hard)
1:15–1:25	Reveal answers, explain Challenge 2 (de-blur)	Check answers, listen
1:25–2:15	Walk around, provide hints	Solve all 3 de-blur videos
2:15–2:25	Reveal answers, explain Challenge 3 (spectrogram)	Check answers, listen
2:25–2:55	Walk around, provide hints	Solve all 3 spectrogram challenges
2:55–3:00	Reveal all answers, final Q&A, distribute prizes	Submit flags (verbally or via Google Form)

After each batch

- Clear score sheet, prepare for next batch (15-min buffer).
- Restart server if needed, restock prizes.

3. What Things You Need

Hardware

- **1 laptop (yours)** – runs HTTP server, has all files and tools.
- **1 backup laptop** – with same files (in case of failure).
- **Projector / large screen** – for demos and reveals.
- **Power strip / extension cord** – for your laptop and participants' laptops.
- **Small table and chairs** – for 10–15 participants.

Software & Files (pre-loaded on your laptop)

- **Python** (to run `python -m http.server`).
- **Challenge files** – 9 files (3 per section) with flags.
- **Portable tools** – FFmpeg, Audacity Portable, GIMP Portable, StegSolve.jar, ExifTool.
- **HTTP server** – started with one command.

Printed Materials

- **Score sheets** (one per batch – participant name, checkboxes for each challenge).
- **Challenge summary sheet** (one per participant – brief instructions, commands).
- **QR code** linking to flag submission form (Google Form) – optional.

Prizes

- Stickers, candy, small tokens – enough for 3 prizes per participant × number of batches × batch size.

4. Budget

Item	Estimated Cost (INR)	Notes
Prizes (stickers/candy)	₹500 – ₹1,000	100–150 items
Printed materials (score sheets, summaries)	₹200 – ₹300	A4 paper, black & white
USB sticks (backup, optional)	₹1,000 – ₹2,000	10–20 sticks
Total	₹1,700 – ₹3,300	(~\$20–\$40 USD)

If the event already provides projector, table, and basic stationery, budget can be as low as ₹500–₹1,000 (prizes only).

5. How Many Volunteers Needed

Role	Number	Responsibilities
Zone Lead (you)	1	Run the 3-hour session, answer technical questions, manage HTTP server, give hints, reveal answers, manage registration check-in.
Assistant (optional)	1	Help distribute prizes, reset score sheets, assist participants who are stuck, manage crowd during batch transition.

If batch size ≤ 10 : You can manage alone.

If batch size 10–15: An assistant is highly recommended.

6. How Many Participants Can Experience Your Zone at a Time

- **Maximum per batch:** 15 participants.
- **Reason:** You can effectively help 15 people during the hands-on phases. More than that becomes chaotic.
- **Total per day:** Depends on number of batches (e.g., 3 batches = 45 participants, 4 batches = 60 participants).

Batch schedule example:

Batch	Time	Capacity
1	9:00 – 12:00	15
2	13:00 – 16:00	15
3	16:00 – 19:00	15
Total daily capacity		45 participants

(Adjust based on event duration.)

. Additional Notes for Organizers

- **Registration required** – no walk-ins. Participants sign up for a time slot.
- **Participants bring their own laptops** with Wi-Fi capability.
- **All tools are portable** – no installation needed. They download from your HTTP server.
- **Internet is optional** – all analysis can be done offline after downloading files.
- **The zone is self-contained** – does not rely on external servers (except optional Google Form for submissions).

